

EXAM 7 – SPRING 2013

19. (6.25 points)

Given the following information for an insurance company:

Loss distribution for <u>earthquake (EQ)</u>		Loss distribution for <u>coastal property (CP)</u>	
<u>Losses (x, in \$000,000)</u>	<u>$f(x)$</u>	<u>Losses (y, in \$000,000)</u>	<u>$g(y)$</u>
0	0.91	0	0.80
200	0.07	250	0.20
400	0.02		

- Losses for these two perils are independent.
- The Wang transform has the form $F^*(x) = \Phi\{\Phi^{-1}[F(x)] - \Phi^{-1}(\alpha)\}$, where Φ represents the cumulative standard normal distribution.
- The Wang transform risk measure for CP is \$197.5 million.
- The Wang transform risk measure for CP and EQ combined is \$323 million.

<u>z</u>	<u>$\Phi(z)$</u>
0.304	0.62
0.409	0.66
0.804	0.79
0.842	0.80
1.341	0.91
1.645	0.95
2.054	0.98

a. (1.75 points)

For EQ, calculate:

- Tail value-at-risk (TVaR) at the 95% level.
- Wang transform at the 95% level.
- Expected policyholder deficit (EPD) at the 5% level.

b. (0.5 point)

Evaluate whether using TVaR as a risk measure for EQ would satisfy the company's requirement to have a pricing charge for any risk taken.

QUESTION 19 CONTINUED ON NEXT PAGE

CONTINUED ON NEXT PAGE

EXAM 7 – SPRING 2013

(19 continued)

c. (0.5 point)

Evaluate whether using the Wang transform as a risk measure for EQ would satisfy the company's requirement to have a pricing charge for any risk taken.

d. (0.5 point)

Evaluate whether using TVaR as a risk measure for EQ would reflect the company's increasing risk aversion to larger losses.

e. (0.5 point)

Evaluate whether using the Wang transform as a risk measure for EQ would reflect the company's increasing risk aversion to larger losses.

f. (2.5 points)

Calculate the TVaR and Wang transform risk measures at the 95% level for the combined CP and EQ book of business. Evaluate whether these measures exhibit the diversification benefits of writing both coverages.

CONTINUED ON NEXT PAGE

Question 19 Sample Answer

Solution 1

- a) $TVaR_{0.95}$ = conditional mean of losses greater than the 95th percentile
 $TVaR_{0.95} = (0.03 \times 200 + 0.02 \times 400) / (0.03 + 0.02) = \mathbf{280M}$

Wang transform

(1)	(2)	(3)	(4)	(5) = (3) - (4)	(6) = $\Phi[(5)]$	
<u>Loss</u>	<u>F(x)</u>	<u>$\Phi^{-1}[F(x)]$</u>	<u>$\Phi^{-1}(\alpha)$</u>		<u>$F^*(x)$</u>	<u>$f^*(x)$</u>
0	0.91	1.341	1.645	-0.304	0.38	0.38
200	0.98	2.054	1.645	0.409	0.66	0.28
400	1.00				1.00	0.34

$$WT(0.95) = (0 \times 0.38) + (200 \times 0.28) + (400 \times 0.34) = \mathbf{192M}$$

$$\begin{aligned} EPD_{\alpha} &= (1-\alpha) \times (TVaR_{\alpha} - VaR_{\alpha}) \\ &= (1-.95) \times (280 - 200) \\ &= \mathbf{4M} \end{aligned}$$

Alternate solution: Some used .05 for α , resulting in **76M**. This was accepted as full credit.

- b) $TVaR_{0.95}$ only includes tail risk and therefore does not meet the criteria to charge for any risk taken.
- c) The Wang transform uses the entire loss distribution to calculate the risk measure and therefore does meet the criteria to charge for all risks taken (not just tail risk).
- d) No. By taking the average of losses above the threshold, TVaR is linear in the tail and fails to demonstrate that the company is more risk averse as the loss size increases.

Alternate solution: Yes. Since it uses tail losses, which would reflect the largest losses of the distribution, it does reflect an increasing risk aversion to larger losses.

- e) Yes. The Wang transform modifies the probability distribution to put more weight on large losses. That is, $f^*(x) > f(x)$ for large losses. This reflects an aversion to larger loss sizes.

f) Assuming independence,

<u>EQ+CP</u>	<u>f(EQ+CP)</u>
0	0.728
200	0.056
250	0.182
400	0.016
450	0.014
650	0.004

$$\begin{aligned}\text{TVaR}_{0.95}(\text{EQ}+\text{CP}) &= (0.016 \times 250) + (0.016 \times 400) + (0.014 \times 450) + (0.004 \times 650) \\ &= \mathbf{386M}\end{aligned}$$

$$\text{TVaR}_{0.95}(\text{EQ}+\text{CP}) < \text{TVaR}_{0.95}(\text{EQ}) + \text{TVaR}_{0.95}(\text{CP})$$

$$386M < 280M + 250M = 530M$$

$\text{TVaR}_{0.95}$ is sub-additive and reflects the diversification benefits.

Wang transform

$$\text{WT}(\text{EQ}) = 192M$$

$$\text{WT}(\text{CP}) = 197.5M$$

$$\text{WT}(\text{EQ}+\text{CP}) = \mathbf{323M}$$

$$\text{WT}(\text{EQ}+\text{CP}) < \text{WT}(\text{EQ}) + \text{WT}(\text{CP})$$

$$323M < 192M + 197.5M = 389.5M$$

Wang transform is sub-additive and reflects the diversification benefits.

For calculation of the Wang Transform, the value was given in the problem. Full credit was awarded to candidates who calculated the number, or for those who pointed out that it was given as 323M.

Examiner Comment

Candidates tended to do well on the TVaR portion of parts a) and f), but not as well on the Wang Transform. They performed best on parts b)-e), even for the parts that focus on the Wang Transform. A common place to lose points was to answer parts b) and c) with a solution that was more applicable to the question asked in parts d) and e).